Where and How Fast Is Heat Stress Worsening? A District-Level, Multi-Index Machine-Learning Projection for 64 districts of Bangladesh

Mohammad Khalid Hasan Nabil[a,*], Mohammad[a], Khalid[a], Hasan Nabil[a], Nabil[a]

[a] *Department of Information Communication Technology, Faculty of Science and Technology, Bangladesh University of Professionals, Dhaka, 1216, Bangladesh*

Abstract

Rising heat and humidity increasingly threaten health, labour, and energy systems across Bangladesh, yet operational planning needs district-resolved, uncertainty-aware projections of the specific conditions that drive human heat stress. We present a deterministic, reproducible machine-learning framework that projects seven temperature-correlated heat-stress conditions heat-index comfort, wet-bulb temperature, indoor and outdoor wet-bulb globe temperature, cooling degree-days, ultraviolet exposure, and dew-point moisture stress for all 64 districts of Bangladesh. Four gradient-boosting ensembles are trained on the 2014–2024 monthly record using future-known seasonal features; the observed 1980–2024 Theil–Sen trend is superimposed so that projections evolve realistically, and all thermal indices are recomputed from the projected primary variables to avoid target leakage. Validated against withheld 2025 observations, the projected indices attain median R^2 of 0.95–0.98, and their operational risk classes match observations

*Corresponding author. ORCID: 0009-0007-3255-1420

Email address: `nabil23549009015@student.bup.edu.bd` (Mohammad Khalid Hasan Nabil)

to within one class in 100% of district-months. A Mann–Kendall analysis shows maximum temperature rising in every district (median +0.56 °C per decade) while minimum temperature remains flat, indicating a widening diurnal range; the coastal and southwestern zones carry the greatest combined burden, and ultraviolet and oppressive-humidity hazards are effectively nationwide. District choropleths, zonal envelopes, and calibrated uncertainty bands provide a directly actionable heat-stress outlook while transparently bounding the models' skill relative to climatology. The complete 64-district, seven-condition projection is openly explorable through an accompanying interactive web application.

Keywords: heat stress, wet-bulb temperature, WBGT, seasonal projection, gradient boosting, Mann–Kendall trend, Bangladesh, climate adaptation

---

1. Introduction

Anthropogenic climate change has fundamentally altered the frequency, intensity, and duration of extreme heat events worldwide. Mean surface air temperatures have risen steadily since the pre-industrial era, and this warming has been accompanied by a disproportionate increase in heatwaves, prolonged humid spells, and compound heat–humidity extremes that the human body increasingly struggles to tolerate [1]. Unlike many climate hazards, extreme heat acts as a silent stressor, producing cumulative physiological strain that is frequently underreported in official mortality and morbidity statistics yet carries substantial public-health, occupational, and economic consequences [2]. Understanding how heat stress manifests across different climatic and geographic settings has therefore become an urgent priority for adaptation planning.

Air temperature alone does not represent the heat stress the human body actually experiences and can lead to misleading assessments. Relative humidity, solar radiation, wind, and atmospheric moisture all govern the body's ability to shed heat through evaporation, radiation, and convection; elevated humidity in particular suppresses the evaporation of sweat the body's primary cooling mechanism so that core temperature rises even when the air is not exceptionally hot [3]. To capture these combined effects, researchers have developed composite heat-stress indices that now underpin public-health warning systems, occupational heat-safety standards [4], and cooling-energy demand projections [5], and that increasingly recognise the approach of physiological survivability limits under extreme humid heat [6].

South Asia is among the regions most acutely exposed to escalating heat stress, owing to its dense population, high monsoon-season humidity, and large share of outdoor and informal-sector labour. Bangladesh occupies a uniquely exposed position within it: a tropical monsoon climate, low-lying deltaic geography, and exceptionally high population density combine with rapid, unplanned urbanization to let heat stress intensify quickly and distribute unevenly across the landscape [7]. Critically, the country is not climatically homogeneous its territory spans drought-prone northwestern plains with continental microclimates, a densely built and heat-retentive central urban core exhibiting strong urban heat-island effects, low-lying coastal districts shaped by maritime humidity, seasonally inundated wetland basins, and elevated hill tracts with distinct topographic regimes. Each zone is expected to experience thermal stress through different pathways and at different intensities, yet most existing heat-risk assessments in Bangladesh continue to treat the country as a single climatic unit or rely on a narrow selection of indices most commonly ambient temperature alone without accounting for the multi-dimensional nature of heat exposure [8].

A parallel gap exists in temporal scope and forecasting capability. Most prior work on heat stress in Bangladesh is retrospective in orientation or resolved only to divisional or national summaries, and very few studies integrate long-term climatological baselines with near-term machine-learning forecasting to project how these selected temperature corrilated conditions will evolve at the district level. This omission matters, because heat-risk management whether through early-warning systems, occupational health policy, or urban cooling infrastructure requires knowing not only where risk is highest today but where and how quickly it is worsening. A recent wet-bulb globe temperature analysis across Bangladesh reported warming of roughly 0.08–0.5 °Cper decade, most pronounced in the southeast and northeast [9], underscoring the need for district-resolved, multi-index projection.

This study addresses these gaps through the Temperature-Correlated Conditions (TCC) framework: an integrated set of seven temperature-dependent indicators heat-index comfort, wet-bulb temperature, indoor and outdoor wet-bulb globe temperature, heatstroke risk, cooling degree-days, ultraviolet exposure, and dew-point moisture stress each capturing a distinct physiological, biophysical, or infrastructural dimension of heat-related risk. Applying this framework to 45 years (1980–2024) of district-level meteorological records across all 64 districts and six climatically distinct zones, we (i) train an ensemble of gradient-boosting models [10] on future-known seasonal features and re-impose the observed long-term trend so that projections evolve realistically rather than collapsing onto the last training year; (ii) recompute every thermal index from the projected primary variables rather than predicting the indices directly, eliminating target leakage and preserving the physical coupling among temperature, humidity, and radiation; (iii) generate 24-month district-level projections (2025–2026) with analytically propagated uncertainty bands; and (iv) validate the projections out-of-sample against independent 2025 observations, benchmark their skill transparently against seasonal-naive and climatological baselines, and characterise long-term change with the non-parametric Mann–Kendall test. The best model per variable is selected by a multi-attribute simple-additive-weighting scheme with a generalization-gap penalty [11].

Two features distinguish the analysis. First, results are reported at full district resolution through choropleth maps rather than aggregated away, exposing spatial gradients such as the coastal concentration of humid-heat burden that zonal summaries obscure. Second, we are explicit about the limits of predictive skill: at monthly resolution the ensembles essentially match climatology, so their value lies in providing spatially complete, uncertainty-aware, and trend-consistent projections of physiologically relevant indices rather than in reducing monthly error. One indicator, heatstroke risk, was found to lack out-of-sample skill and is retained only descriptively. The remainder of this paper details the data and methods, presents the validated projections and observed trends, and discusses their implications for heat-adaptation planning across Bangladesh.

2. Literature Review

Human exposure to heat is not adequately captured by dry-bulb air temperature alone. A growing body of work therefore evaluates compound thermal indices that combine temperature with humidity, radiation, and wind to describe how heat is actually experienced and how it threatens health, labour productivity, and energy demand. This section synthesises recent international and Bangladesh-focused literature across the seven temperature-correlated conditions addressed in this study, the machine-learning methods used to forecast them, and the urban and regional context that motivates a district-resolved, multi-index treatment.

2.1. The physiological basis of thermal-stress indices

Raymond et al. [6] provided seminal evidence that a wet-bulb temperature (WBT) of 35 °C represents the physiological limit for human survival, showing that humid-heat extremes had already more than doubled in frequency between 1979 and 2017 and shifting the field's focus from ambient temperature toward compound heat–humidity metrics. The inadequacy of dry-bulb temperature as a standalone indicator is well established: high humidity impairs the evaporative efficiency of perspiration, causing perceived loads to exceed measured air temperature [3]. The heat index formalises this through the regression of Rothfusz [12], operationalised by the U.S. National Weather Service. The wet-bulb globe temperature (WBGT), originally developed by Yaglou and Minard [13] for military training and codified in ISO 7243 [4], extends the framework to radiant load, making it the international standard for occupational heat-stress assessment [14]. The empirical WBT approximation of Stull [15], derived by gene-expression programming with a mean absolute error below 0.3 °C, provides a computationally efficient estimate applicable to station and reanalysis data alike. Cooling degree-days (CDD) complement these physiological indices by quantifying the cumulative thermal energy above a baseline, a standard proxy for building cooling demand [5, 16]; the UV index [17] captures the solar-radiation dimension of outdoor exposure; and dew-point temperature offers a stable, temperature-independent measure of atmospheric moisture that outperforms relative humidity as a predictor of perceived discomfort [18]. Together these conditions span the physiological, biophysical, and infrastructural dimensions of heat risk that the present study projects jointly, against a backdrop of accelerating global warming documented by the IPCC [19].

2.2. Observed heat-stress trends in Bangladesh

Bangladesh is among the most heat-exposed nations, and a dense body of station-based work documents its warming. Rahman et al. [20] produced the first comprehensive spatiotemporal appraisal of historical and projected heat-index change, finding statistically significant upward trends across most districts, sharpest in the pre-monsoon season. Ekra et al. [21] analysed 60 years (1961–2020) of records from 17 stations using the Discomfort Index, Humidex, and WBT, reporting a spatially uneven rise in heat discomfort that is steepest along the coastal belt and driven primarily by rising air temperature. For WBGT specifically, Kamal et al. [9] provided the anchor analysis, computing outdoor WBGT from 1979–2021 reanalysis with the physically based Liljegren model and finding a national rise of 0.08–0.5 °C per decade, a fivefold expansion of the area affected by high-to-extreme stress, and air temperature as the dominant driver (85.9% relative importance); a companion paper developed simplified regression equations to lower the computational barrier to WBGT adoption [22]. Temperature-extreme indices are likewise intensifying: Ahmed et al. [23] found lengthening warm-spell durations, particularly in coastal regions, and Islam et al. [24] identified warm-index trends dominated by post-2000 increases across 27 stations. Trends in degree-days were examined by Islam et al. [25], who reported significant CDD increases with the southwestern and central zones showing the highest annual means. At the health end, Miah et al. [26] linked ambient temperature and heat index to heatstroke mortality in an ecological time-series analysis, reinforcing the public-health urgency of a multi-index approach.

2.3. Urban heat islands and zone-specific dynamics

Urban heat-island (UHI) intensity amplifies background warming, especially within the densely built central zone examined here. Dewan et al. [27] characterised diurnal and seasonal UHI across five major Bangladeshi cities from satellite land-surface temperature, finding the greatest magnitude in the pre-monsoon period and the largest daytime surface UHI in Dhaka (2.74 °C), followed by Chittagong (1.92 °C) and Khulna (1.27 °C). Faisal et al. [28] reported that UHI intensity rose substantially across most districts from 2000 to 2020, with Dhaka and Mymensingh steepest, and Park et al. [29] showed that Dhaka's UHI is strongest in winter (0.95 °C) and weakest during the monsoon (0.23 °C), with heat waves producing distinctive synoptic patterns that compound local warming. These findings support treating Bangladesh's urban core as a distinct climatic zone and, more broadly, motivate the six-zone structure adopted in this study.

2.4. Machine learning for meteorological and thermal-index forecasting

Ensemble tree methods have consistently outperformed conventional statistical and numerical approaches for temperature and humidity forecasting in data-rich, strongly seasonal settings. The gradient-boosting family XGBoost [30], LightGBM [31], and CatBoost [32] recurs as a strong, interpretable baseline across environmental prediction problems. In Bangladesh, Mohsin et al. [33] found XGBoost most accurate for seasonal temperature prediction across 29 stations ($R^2 = 0.88$ for maximum and 0.97 for minimum temperature) when trained on ENSO predictors; Akter and Rahman [34] found Random Forest best for humidity forecasting at northern stations over a 40-year record; and Rahman et al. [35] found a stacking ensemble outperformed RNN–LSTM and decision-forest models for average-temperature prediction over a 60-year series. Most directly, Binte Ahmed et al. [36] benchmarked ARIMAX, SARIMAX, Random Forest, XGBoost, and LSTM on 3-hourly Dhaka records (2014–2023) for conditional, scenario-based heat-index projection, with Random Forest achieving the lowest error (RMSE = 0.85 °C, $R^2 = 0.987$) a close methodological precedent for the present work. The value of combining gradient-boosting learners is illustrated by Kim et al. [37], whose stacked XGBoost/LightGBM/CatBoost ensemble reached $R^2 = 0.93$, outperforming any single model. Fourier-based encoding of seasonal cycles as sine and cosine terms has been shown to improve both accuracy and efficiency on meteorological series [38], directly motivating the Fourier feature layer used here. Comparative studies elsewhere reinforce the transferability of these methods [39, 40], while the validation of deep-learning weather models on recent extreme events [41] and analyses of the distinct atmospheric drivers of humidex versus temperature heatwaves [42] situate the forecasting problem in its broader physical context.

2.5. Per-condition modelling and international context

Beyond the anchor WBGT and heat-index studies, the individual conditions have distinct literatures. For the heat-index/Humidex family, Sojan and Srinivasan [43] showed that humid-heat stress in India is intensifying disproportionately faster than dry heat; Kushwaha et al. [44] projected air temperature as the dominant driver of future extreme-Humidex days across 15 CMIP6 models; Scoccimarro et al. [45] established that perceived-temperature extremes intensify faster than temperature extremes owing to rising humidity; Saha et al. [46] documented a post-2000 escalation of humid heatwaves across Northeast India and Bangladesh; and Tabassum et al. [47] connected a cool-roof intervention to measurable Humidex reductions during a Dhaka heat wave. Heatstroke-risk modelling has matured chiefly outside Bangladesh Xu et al. [48] trained interpretable models linking meteorology to heatstroke incidence in South China with Ehsan et al. [49] providing a recent Bangladesh-focused machine-learning forecast of extreme-heat health impact; consistent with this thin evidence base, the present study computes a heatstroke indicator but, as reported below, excludes it from the validated results for a lack of out-of-sample skill. For CDD, Bilgili [50] showed that classical SARIMA models remain competitive with machine learning when the seasonal structure is strong. UV-index modelling is comparatively underexplored in the regional literature: Teggi et al. [51] recently derived the UV index from reanalysis validated against ground stations, whereas in the present framework the UV index is instead predicted directly as a modelled variable. Dew-point prediction has a well-developed machine-learning lineage, progressing from extreme-learning-machine and kernel baselines [52, 53], to gradient-boosted trees exploiting cross-station information [54], to attention-based sequence models with explainability [55]. Internationally, the stakes are large: Parsons et al. [56] estimated humid-heat labour losses equivalent to roughly 1.7% of 2017 global GDP, with South Asian outdoor workers among the worst affected, and Ioannou et al. [57] meta-analysed a significant negative association between WBGT exposure and manual work capacity. Longer-term projections for Bangladesh under CMIP6 [58] indicate continued rises in warm-spell duration and tropical nights, providing the multi-decadal backdrop against which the near-term (2025–2026) forecasts generated here should be read.

2.6. Research gaps addressed by this study

Three gaps emerge from this review. First, although individual indices chiefly WBGT and the heat index have been examined for Bangladesh in isolation, no study has simultaneously computed, validated, and forecast seven complementary thermal-stress conditions within a single framework, at district resolution, across climatically distinct zones. Second, existing Bangladeshi forecasting work has relied on conventional statistics, CMIP6 downscaling, or single-algorithm machine learning; the multi-algorithm gradient-boosting ensemble with Fourier feature engineering and an explicitly re-imposed observational trend proposed here has not previously been applied to thermal-stress forecasting in this context. Third, the six-zone climatological structure of Bangladesh Northwest dry-hot plains, Central urban core, Southwest saline-drought belt, Coastal maritime zone, Haor wetland basins, and Hill Tracts has not previously organised a multi-index heat-risk assessment, leaving intra-national thermal heterogeneity systematically undercharacterised. In contrast to studies built on ERA5 reanalysis, the present work is driven by station-derived Visual Crossing records and validated out-of-sample against independent 2025 observations, providing an externally verified, uncertainty-aware, and district-resolved projection across all seven conditions.

3. Materials and Methods

This study builds a deterministic, reproducible pipeline that (i) learns the monthly climatology of seven primary meteorological variables for each of the 64 districts of Bangladesh from a recent, high-quality window; (ii) superimposes the long-term observed climate trend so that projections evolve realistically beyond the training period; (iii) recomputes physiologically relevant thermal indices from the projected primary variables to avoid target leakage; (iv) quantifies projection uncertainty by first-order error propagation; and (v) validates the projected 2025 fields against independent observations. All methodological constants are fixed a priori and are reported here so that the workflow can be audited and reproduced exactly.

3.1. Study area and climatic zonation

The analysis covers all 64 administrative districts of Bangladesh. For synthesis and reporting, districts are grouped into six climatic zones: (1) Northwest Extreme Heat Zone (dry hot plains, including the Barind Tract), (2) Central Urban Heat Zone (urban heat-island region), (3) Southwest Hot–Dry Zone (drought- and salinity-affected areas), (4) Coastal Humid Heat Zone (the coastal belt), (5) Haor/Wetland Humid Zone (the haor basin), and (6) Hill Tract Zone (elevated terrain). The zonation is author-defined but grounded in converging, established frameworks: the agro-climatic regionalisation of the Bangladesh Meteorological Department and the agro-ecological/physiographic zonation of Bangladesh [59, 60], together with recognised hydro-ecological units (Barind Tract, Sundarbans coastal belt, haor basin). Each district is flagged as a *confirmed* (high-confidence) or *provisional* assignment, and one representative *exemplar* district per zone is retained for main-text figures (Figure 1). The complete district-to-zone mapping is provided in the supplementary material.

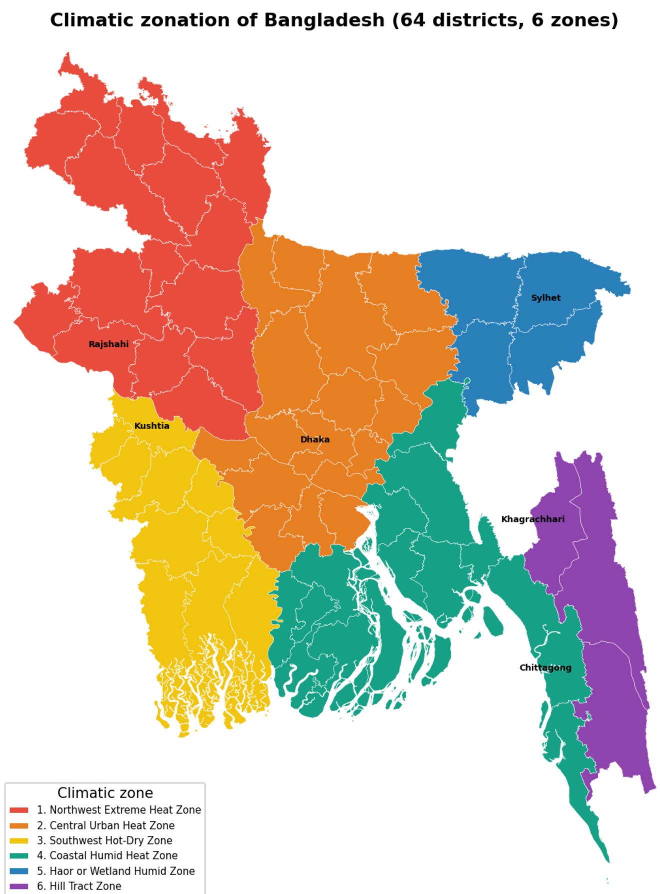

Figure 1: Climatic zonation of Bangladesh: the 64 districts grouped into six zones, with the exemplar district of each zone labelled.

3.2. Data and preprocessing

Daily meteorological records were obtained from the Visual Crossing Weather API [61] for each district. Seven primary (physically measured) variables are used: mean, maximum, and minimum near-surface air temperature (°C); relative humidity (%); dew-point temperature (°C); downward solar (shortwave) radiation; and the ultraviolet (UV) index. The historical record spans 1980–2024; an additional file covering 2025 (through 20 November 2025; the final month is partial) is reserved exclusively for out-of-sample validation.

Because humidity is noisy in the earliest decades and solar/UV are unavailable before 2014, the machine-learning window is restricted to 2014–2024, whereas the full 1980–2024 record is used only for the observational trend analysis (Section 3.12). Daily values are aggregated to calendar-month means (month-start resampling), and residual gaps are filled by linear interpolation. All modelling is performed on these monthly series.

3.3. Feature construction

To respect a strict *future-known-features-only* constraint no lags, rolling windows, or autoregressive terms that would be unavailable at projection time each month is described by a deterministic seasonal–secular feature vector. Seasonality is encoded by a truncated Fourier series with period $P = 12$ months and $H = 3$ harmonics,

$$\sin_h(m) = \sin\left(\frac{2\pi hm}{P}\right), \qquad \cos_h(m) = \cos\left(\frac{2\pi hm}{P}\right), \qquad h = 1, \ldots, H, \quad (1)$$

where m is the calendar-month index. The secular component is the calendar year. The full design vector is therefore $\mathbf{x} = [\,\text{year},\ \sin_1, \cos_1, \ldots, \sin_3, \cos_3\,]$ (seven features).

3.4. Predictive models

Four gradient-based tree ensembles are trained independently for every district–variable pair: Random Forest [10], XGBoost [30], LightGBM [31], and CatBoost [32]. Hyperparameters are fixed a priori and applied uniformly across all districts and variables so that results are mutually comparable and the pipeline is fully deterministic (global random seed = 42). The fixed configurations are listed in the supplementary material.

3.5. Trend-aware detrending and projection

Tree ensembles cannot extrapolate beyond the range of their training features; if the calendar year is used directly as a predictor, every projected year collapses onto the last training year, producing a static (trendless) forecast. To carry the observed climate trend forward while preserving the future-known constraint, each primary variable is detrended before learning and re-trended at prediction time. For district–variable series $y(t)$ with continuous time t (in years), a Theil–Sen slope $\hat{\beta}$ is estimated from the annual-mean series,

$$\hat{\beta} = \text{median}_{i<j} \frac{\bar{y}_j - \bar{y}_i}{t_j - t_i}, \tag{2}$$

where $\bar{y}_k$ is the mean of year k. To keep the projected trend consistent with the manuscript's observational trend analysis, $\hat{\beta}$ is taken from the robust 1980–2024 record for the five variables that span it (mean/max/min temperature, humidity, dew point); solar radiation and the UV index, which lack a pre-2014 record, fall back to a Theil–Sen fit over their available 2014–2024 window. The ensembles are trained on the deseasonalised residual $r(t) = y(t) - \hat{\beta}\, t$, so they model only the seasonal signal; the linear trend is added back at prediction time, $\hat{y}(t) = \hat{r}(t) + \hat{\beta}\, t$. This design makes each projected year differ from the previous one by exactly $\hat{\beta}$ and guarantees that the projected inter-annual tendency equals the reported Mann–Kendall/Sen slope (Section 3.12). We emphasise that this trend term is included for physical consistency with the observed record, not to improve short-horizon accuracy: at monthly resolution the seasonal cycle dominates, so the trend contributes only a small secular adjustment.

The winning model per variable (Section 3.6) is refit on the full 2014–2024 residual and projected forward for a 24-month horizon (January 2025 through December 2026). Projected primary variables are clipped to physically plausible bounds before any downstream computation to prevent non-physical extrapolations from propagating into the indices.

3.6. Model evaluation and selection

Reported test metrics use a strictly chronological hold-out: models are trained on 2014–2022 and tested on the withheld 2023–2024 period. Generalisation is additionally assessed by five-fold `TimeSeriesSplit` cross-validation on the residual series. For each candidate model we record the coefficient of determination R^2 and root-mean-square error (RMSE) on the chronological test set, the cross-validated R^2 (CV-R^2), a generalisation gap Gap $=$ $|R^2 - \text{CV-}R^2|$, and an operational tolerance accuracy A_τ defined as the fraction of test months whose absolute error does not exceed a variable-specific tolerance τ,

$$A_\tau = \frac{1}{N}\sum_{n=1}^{N} \mathbf{1}\big(|y_n - \hat{y}_n| \le \tau\big). \tag{3}$$

The winning model is chosen by Simple Additive Weighting (SAW), a multi-attribute decision-making scheme [11], with a generalisation-gap penalty. Each criterion c is min–max normalised across the four candidates directly for benefit criteria and by complement for cost criteria,

$$\tilde{s}_c = \begin{cases} \dfrac{s_c - \min_c}{\max_c - \min_c}, & c \text{ maximised}, \\[2ex] \dfrac{\max_c - s_c}{\max_c - \min_c}, & c \text{ minimised}, \end{cases} \tag{4}$$

and the composite score is the weighted sum $S = \sum_c w_c\, \tilde{s}_c$, with weights $w_{R^2} = 0.25$, $w_{\text{CV-}R^2} = 0.25$, $w_{\text{RMSE}} = 0.20$, $w_{\text{Gap}} = 0.20$, and $w_{A_\tau} = 0.10$ (RMSE and Gap treated as cost criteria). The highest-scoring model wins; models with Gap > 0.10 are additionally flagged as overfitting risks.

3.7. Thermal index computation

To avoid target leakage, thermal-stress indices are never predicted directly; they are recomputed analytically from the *projected* primary variables. The heat index (HI) uses the U.S. National Weather Service Rothfusz regression [12]. With temperature T_F and relative humidity R (in °F and %),

$$\begin{aligned}
\text{HI} = \; & -42.379 + 2.04901523\, T_F + 10.14333127\, R - 0.22475541\, T_F R \\
& - 6.83783\times 10^{-3} T_F^2 - 5.481717\times 10^{-2} R^2 + 1.22874\times 10^{-3} T_F^2 R \\
& + 8.5282\times 10^{-4} T_F R^2 - 1.99\times 10^{-6} T_F^2 R^2,
\end{aligned} \tag{5}$$

with the standard low-humidity and high-humidity adjustments applied within their validity ranges and a simplified formula substituted below 80°F; the result is returned in °C. Wet-bulb temperature T_w follows the empirical formulation of Stull [15],

$$\begin{aligned}
T_w = T \,\arctan\!\left[0.151977\, (R + 8.313659)^{1/2}\right] + \arctan(T + R) \\
- \arctan(R - 1.676331) + 0.00391838\, R^{3/2} \arctan(0.023101\, R) - 4.686035,
\end{aligned} \tag{6}$$

with T in °C and R in %. Wet-bulb globe temperature is computed in a shade variant (no direct-beam term) and a sun variant that adds a solar-radiation contribution S,

$$\mathrm{WBGT}_{\mathrm{shade}} = 0.7\,T_w + 0.3\,T, \tag{7}$$

$$\mathrm{WBGT}_{\mathrm{sun}} = 0.7\,T_w + 0.2\,(T + 0.025\,S) + 0.1\,T. \tag{8}$$

Cooling degree-days are expressed as a monthly-mean daily rate about an 18 °C base,

$$\mathrm{CDD} = \max\Big(\tfrac{T_{\max}+T_{\min}}{2} - 18,\ 0\Big). \tag{9}$$

A composite heatstroke-hazard score (a sigmoid of the heat index modulated by a wet-bulb multiplier) was also implemented and recalibrated for monthly-mean inputs. However, in external validation it produced a negative out-of-sample R^2 (i.e. worse than a constant predictor), so it is *excluded from all predictive results* and retained only as a descriptive projected field. This exclusion is stated explicitly to avoid over-interpretation of an index that does not demonstrate skill.

3.8. Operational heat-stress conditions and risk categorisation

The projected fields are organised into seven temperature-correlated heat-stress conditions of operational relevance: (1) heat-index comfort (perceived temperature under humidity); (2) wet-bulb temperature (the thermodynamic limit of evaporative cooling); (3) wet-bulb globe temperature in indoor (shade) and outdoor (sun) forms (the occupational heat-exposure standard); (4) heatstroke risk; (5) cooling degree-days (a building-cooling energy proxy); (6) ultraviolet exposure risk; and (7) dew-point moisture stress. Conditions (6) and (7) are obtained directly from the projected UV-index and dew-point fields, which are among the primary modelled variables (Section 3.2); condition (4) is excluded from predictive results as noted above.

To move from continuous values to operational meaning, four of the conditions are additionally mapped to established risk-category systems: the U.S. National Weather Service heat-index comfort classes (Comfortable, Caution, Extreme Caution, Danger, Extreme Danger); WBGT occupational-exposure classes (Low, Moderate, High, Extreme); the World Health Organization UV-index classes (Low, Moderate, High, Very High, Extreme) [17]; and dew-point human-comfort classes (Dry through Extremely Oppressive). Wet-bulb temperature (2) and cooling degree-days (5) are reported as continuous physical quantities. The category boundaries are conventionally defined for daily peak values; because the present pipeline operates on monthly means, the extreme classes are seldom reached and the categories should be read as describing *typical monthly* conditions rather than daily peaks. Categorisation skill is assessed against the observed 2025 fields by exact-match agreement and by agreement to within one adjacent class.

3.9. Uncertainty quantification

Projection uncertainty for the temperature–humidity indices is obtained by first-order (linearised) error propagation from the primary-variable uncertainties. For an index $I(x_1, \dots, x_k)$ with input standard deviations σ_{x_j},

$$\sigma_I = \left[\sum_j \left(\frac{\partial I}{\partial x_j} \right)^2 \sigma_{x_j}^2 \right]^{1/2}, \tag{10}$$

where the input σ_{x_j} are taken from each winning model's in-sample RMSE and representative sensitivity coefficients are used for the heat index, wet-bulb temperature, and WBGT. The propagated σ_I define the 68% ($\pm\sigma_I$) and 95% ($\pm 1.96\,\sigma_I$) bands reported for the projected indices.

3.10. Baseline skill benchmarks

Model skill is benchmarked against two references evaluated on the same chronological test set: a seasonal-naive predictor (each test month set to the same calendar month one year earlier) and a monthly-climatology predictor (the training-period mean for each calendar month). Skill relative to a reference is

$$\mathrm{SS} = 1 - \frac{\mathrm{RMSE}_{\mathrm{model}}}{\mathrm{RMSE}_{\mathrm{ref}}}, \tag{11}$$

so that positive values indicate improvement over the reference.

3.11. External validation against observed 2025

The 24-month projection includes calendar year 2025, for which independent observations were withheld from all training. Projected monthly primary variables and recomputed indices are compared with the observed 2025 monthly series using R^2, RMSE, mean absolute error, and the tolerance-accuracy bands of Section 3.6. The final month (November 2025) is partial in the source data and is treated accordingly. This constitutes a genuine, fully out-of-sample test of the projection.

3.12. Observational trend analysis

Long-term trends are characterised on the full 1980–2024 annual-mean series using the non-parametric Mann–Kendall test with Theil–Sen slope estimation, computed independently of the predictive models (and therefore free of target leakage). The Mann–Kendall statistic is

$$S = \sum_{k=1}^{n-1} \sum_{j=k+1}^{n} \mathrm{sgn}(x_j - x_k), \tag{12}$$

with tie-corrected variance

$$\mathrm{Var}(S) = \frac{n(n-1)(2n+5) - \sum_i t_i(t_i-1)(2t_i+5)}{18}, \tag{13}$$

where t_i is the number of ties of extent i. The standardised statistic

$$Z = \begin{cases} (S-1)/\sqrt{\mathrm{Var}(S)}, & S > 0, \\ 0, & S = 0, \\ (S+1)/\sqrt{\mathrm{Var}(S)}, & S < 0, \end{cases} \tag{14}$$

is referred to the standard normal distribution for a two-sided p-value, and Kendall's $\tau = 2S/[n(n-1)]$ measures rank correlation with time. The Theil–Sen slope (Eq. 2) provides the trend magnitude, reported per decade. Trends with $p < 0.05$ are deemed significant. This analysis is performed for mean, maximum, and minimum temperature, humidity, dew point, and the temperature–humidity indices (heat index, wet-bulb temperature, shade WBGT) that are computable across the full record.

A subset of districts encode pre-$\sim$2000 missing days as zero in the source record; left uncorrected, the resulting $0 \to\sim 30\,^{\circ}\mathrm{C}$ step fabricates physically impossible trends (an artefactual $+7.8\,^{\circ}\mathrm{C}$ per decade in one district). Before any trend is fit, daily values below physically plausible floors ($5\,^{\circ}\mathrm{C}$ for temperatures, with lower floors for minimum temperature and dew point, and 5% for humidity) are therefore treated as missing, and a year must retain at least 60 valid days to enter the annual-mean series. The same masked slopes are used when the trend is re-added to the projection (Section 3.5). This affects only the observational trend and long-run slope; the machine-learning models are unaffected because they train exclusively on the complete 2014–2024 window.

3.13. Zonal synthesis

District-level projections are summarised by zone as monthly and annual envelopes (the minimum, mean, and maximum across member districts) for the key indices, providing compact, journal-page-efficient views of the projected heat-stress distribution in place of 64 individual district panels. Full district-level detail is additionally retained through choropleth maps rendered directly from the CC BY 3.0 IGO geoBoundaries Bangladesh ADM2 administrative boundaries [62]; districts are matched to the boundary geometry by name, and per-district values are mapped for the observed trends, the projected annual means of all seven conditions, and the number of months per year each district spends in each condition's hazard class.

3.14. Implementation and reproducibility

The pipeline is implemented in Python 3.14 using scikit-learn 1.8 [63], LightGBM 4.6, XGBoost 3.3, CatBoost 1.2.10, together with NumPy, pandas, and Matplotlib. All stochastic components use a fixed seed, and a run manifest records package versions, configuration constants, and per-run timings so that every reported artefact is traceable to a specific configuration. The complete source code, the reviewer workbook, and an interactive web application that exposes every district and condition are released with the study (see Data availability).

4. Results

The full pipeline was executed for all 64 districts, producing 448 district–variable models, a 24-month projection (January 2025–December 2026) per district, an out-of-sample validation against observed 2025, and the 1980–2024 observational trend analysis. Unless stated otherwise, statistics are reported as the median across the 64 districts. Because the complete per-district projections, validations, and risk categories for all seven conditions cannot be reproduced in print, they are additionally provided through an interactive web dashboard[1] through which any district and condition can be explored directly.

4.1. Predictive accuracy and model selection

On the chronological 2023–2024 hold-out, the ensembles reproduced the primary variables accurately, with median test R^2 of 0.970 for minimum temperature, 0.961 for mean temperature, 0.962 for dew point, and 0.886 for maximum temperature; humidity (0.632), solar radiation (0.691), and the UV index (0.664) were intrinsically harder and scored lower. Under the SAW criterion, LightGBM was selected most frequently (284 of 448 fits), followed by Random Forest (111) and CatBoost (53); XGBoost, although included in the candidate pool, was never top-ranked. Model preference was variable-dependent (Table 1): LightGBM dominated temperature (57/64), maximum temperature (63/64), and dew point (48/64), whereas Random Forest was preferred for the noisier humidity field (55/64) and was competitive for solar radiation and UV.

4.2. External validation against observed 2025

The projected 2025 fields, produced without any exposure to 2025 observations, validated strongly (Table 2). Median R^2 reached 0.982 for minimum temperature, 0.980 for dew point, and 0.950 for mean temperature (RMSE 0.72 °C). Crucially, the thermal-stress indices recomputed from the projected primary variables rather than predicted directly also validated well: wet-bulb temperature and shade WBGT both at $R^2 = 0.969$ (RMSE ≤ 0.63 °C), sun WBGT at 0.966, the heat index at 0.951 (RMSE 1.36 °C), and cooling degree-days at 0.951. Maximum temperature (0.845), humidity

[1] `https://tcc-64-districts-mohammad-khalid.streamlit.app/`

Table 1: Best-model selection frequency across the 64 districts, by variable (SAW criterion). XGBoost was never selected and is omitted.

Variable	LightGBM	Random Forest	CatBoost
Mean temperature	57	0	7
Maximum temperature	63	0	1
Minimum temperature	44	2	18
Relative humidity	4	55	5
Dew point	48	0	16
Solar radiation	29	29	6
UV index	39	25	0

(0.825), UV (0.805), and solar radiation (0.766) were the weakest, consistent with their lower learnability. For the six exemplar districts, heat-index validation R^2 ranged from 0.873 (Sylhet) and 0.891 (Chittagong) to 0.948–0.954 (Kushtia, Khagrachhari, Rajshahi, Dhaka), with RMSE between 1.1 and 1.9 °C.

4.3. Skill relative to baselines

Against the seasonal-naive reference the ensembles were skilful in 81.5% of district–variable cases, with the largest median gains for minimum temperature (+0.24) and dew point (+0.21). Relative to monthly climatology, however, the models were essentially at parity: the median skill score was near zero or slightly negative for every variable (e.g. −0.003 for mean temperature, −0.030 for maximum temperature, −0.059 for minimum temperature), and only 31.9% of cases exceeded climatology. This is the expected behaviour at monthly resolution: the ensembles reconstruct the strong tropical seasonal cycle that monthly climatology already encodes, so they add

Table 2: External validation of the projected 2025 fields against observed 2025, summarised as the median across 64 districts. Errors are in each variable's native unit (°C for temperatures and thermal indices; % for humidity; $\mathrm{W\,m^{-2}}$ for solar radiation; index units for UV; $^\circ\mathrm{C\,d\,d^{-1}}$ for CDD).

Variable	n	Median R^2	Median RMSE	Median MAE
Mean temperature	64	0.950	0.720	0.615
Maximum temperature	64	0.845	0.940	0.788
Minimum temperature	64	0.982	0.565	0.449
Relative humidity	64	0.825	2.334	1.996
Dew point	64	0.980	0.601	0.469
Solar radiation	64	0.766	14.352	11.704
UV index	64	0.805	0.375	0.302
Heat index	64	0.951	1.357	1.142
Wet-bulb temperature	64	0.969	0.624	0.509
WBGT (shade)	64	0.969	0.595	0.506
WBGT (sun)	64	0.966	0.644	0.543
Cooling degree-days	64	0.951	0.696	0.577

little in raw error terms. Their contribution is therefore not a reduction in monthly RMSE but a spatially complete, uncertainty-quantified, and trend-consistent projection that climatology cannot provide.

4.4. Observed long-term trends, 1980–2024

The Mann–Kendall/Sen analysis revealed a coherent warming signal (Table 3, Figure 2). Maximum temperature increased significantly in all 64 districts (median +0.56 °C per decade), and the heat index, wet-bulb temperature, and shade WBGT increased in 58, 57, and 58 of 64 districts (median +0.37, +0.17, and +0.17 °C per decade, respectively). Mean temperature rose in 50 districts and dew point in 46. Minimum temperature was markedly more heterogeneous (16 increasing, 13 decreasing, 35 without significant trend; median +0.02 °C per decade), and humidity was mixed (24 increasing, 23 decreasing). The sharp contrast between near-universal maximum-temperature warming and the essentially flat minimum-temperature signal indicates a widening diurnal temperature range in much of the country; a small number of districts (e.g. Sylhet) therefore show a slight decrease in *mean* temperature despite rising maxima. Because the projection re-imposes exactly these Sen slopes, the projected inter-annual tendency reproduces this observed signal by construction.

4.5. Projected 2025–2026 outlook and zonal synthesis

Consistent with the modest observed slopes, the projected year-over-year change was small: the heat index rose by a median of +0.038 °C per year across the 64 districts (range −0.034 to +0.096 °C per year), the negative values corresponding to the districts whose 1980–2024 mean-temperature trend is itself slightly negative. The zonal synthesis (Table 4) places the

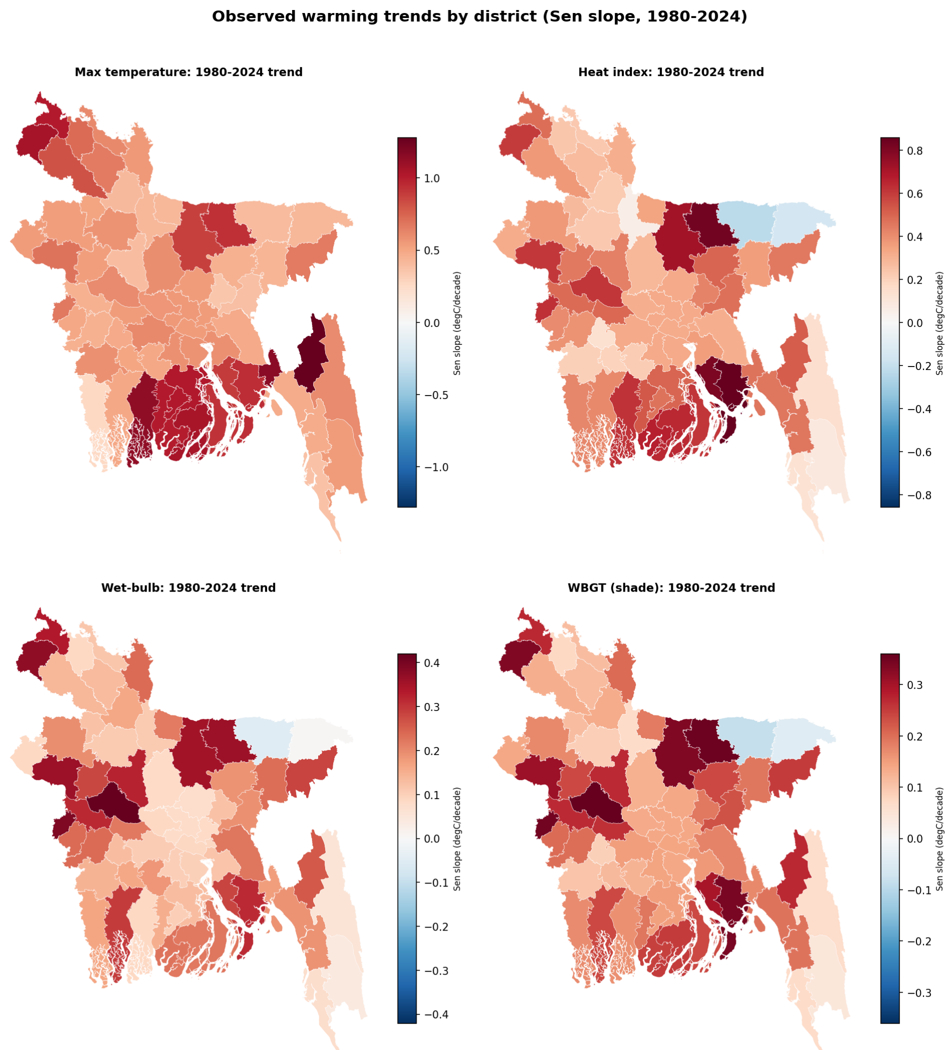

Figure 2: Observed 1980–2024 warming by district (Theil–Sen slope, °C per decade) for maximum temperature, heat index, wet-bulb temperature, and shade WBGT. Red indicates warming, blue cooling. Values are computed after masking missing-as-zero early-record days (Section 3.12).

Table 3: Observed 1980–2024 trends (Mann–Kendall test with Theil–Sen slope) by variable: number of districts increasing, decreasing, or without a significant trend ($p < 0.05$), and the median Sen slope per decade (°C per decade, except humidity in % per decade).

Variable	Increasing	Decreasing	No trend	Median slope/decade
Maximum temperature	64	0	0	+0.558
Mean temperature	50	2	12	+0.194
Minimum temperature	16	13	35	+0.021
Dew point	46	0	18	+0.197
Relative humidity	24	23	17	+0.177
Heat index	58	1	5	+0.373
Wet-bulb temperature	57	0	7	+0.165
WBGT (shade)	58	0	6	+0.172

highest projected annual-mean heat index in the Southwest Hot–Dry Zone (31.0 °C) and the Coastal Humid Heat Zone (30.6 °C), and the lowest in the Haor/Wetland Zone (28.0 °C); sun WBGT peaked in the coastal belt (25.5 °C). The two continuous-burden conditions reinforce this ordering: projected cooling degree-days a proxy for building cooling-energy demand were highest in the Southwest Hot–Dry and Coastal Humid Heat zones (9.1 and 9.0 $^{\circ}\mathrm{C\,d\,d^{-1}}$) and lowest in the Haor/Wetland Zone (7.0), while projected wet-bulb temperature, the thermodynamic ceiling on evaporative cooling, followed the same spatial pattern, peaking in the Coastal (23.7 °C) and Southwest (23.6 °C) zones and again reaching its minimum in the Haor basin (22.6 °C). The coastal belt therefore emerges as the zone of greatest combined perceived-heat, humid-heat, and cooling-energy burden. Figure 6 maps all six continuous conditions at district resolution, confirming the coastal-south concentration of heat, humidity, and cooling demand. Figure 3 illustrates the full record–validation–projection narrative for the Dhaka exemplar, and Figure 4 summarises the projected monthly climatology of all six key conditions across the six zones, exposing their distinct seasonal signatures. Figure 5 shows the district spread within the hottest (Southwest) zone.

Table 4: Projected annual-mean thermal indices by climatic zone (zone mean, with the across-district minimum and maximum in parentheses). Values in °C except CDD ($^\circ\mathrm{C\,d\,d^{-1}}$).

Zone	Heat index	WBGT (sun)	Wet-bulb	CDD
1. Northwest Extreme Heat	29.6 (28.4–30.7)	24.7	22.9	8.3
2. Central Urban Heat	29.6 (29.3–30.9)	24.8	22.8	8.8
3. Southwest Hot–Dry	31.0 (30.6–31.9)	25.5	23.6	9.1
4. Coastal Humid Heat	30.6 (29.5–31.5)	25.5	23.7	9.0
5. Haor/Wetland Humid	28.0 (26.0–29.5)	24.3	22.6	7.0
6. Hill Tract	29.9 (29.3–30.3)	25.4	23.5	8.6

4.6. Operational risk categories

Mapping the projected fields onto their risk-category systems yields a policy-facing summary of the seven conditions (Table 5). Under monthly-mean heat index, projected months split between Comfortable (37.8%) and Extreme Caution (51.8%), with the intermediate Caution class at 10.2% and the Danger class essentially absent (0.1%) the latter a direct consequence of categorising monthly means rather than daily peaks. Outdoor WBGT was predominantly Low (59.0%) to Moderate (37.0%), reaching the High class in 3.9% of months, while indoor (shade) WBGT rarely left the

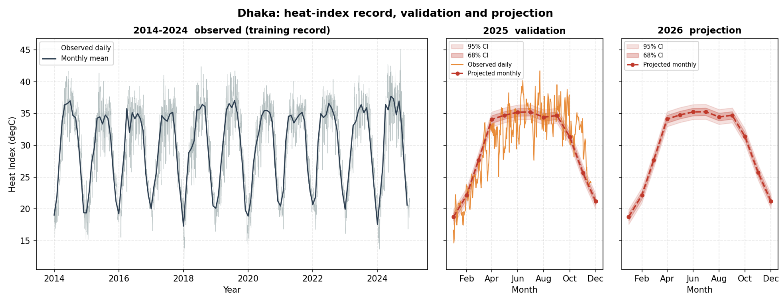

Figure 3: Heat-index record, validation, and projection for the Dhaka exemplar: (left) observed daily heat index 2014–2024 with the monthly mean; (centre) the 2025 validation year, observed daily versus the monthly projection with 68% and 95% confidence bands; (right) the 2026 monthly projection with confidence cloud.

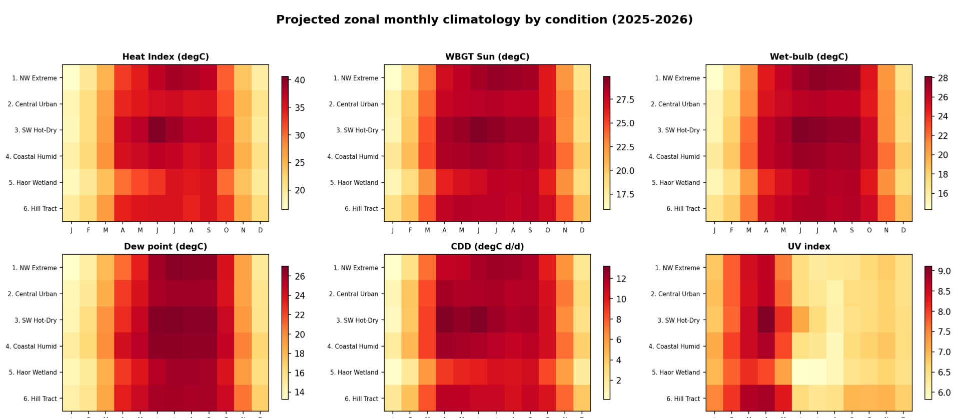

Figure 4: Projected zonal monthly climatology for six heat-stress conditions (heat index, sun WBGT, wet-bulb temperature, dew point, cooling degree-days, and UV index). The temperature-driven conditions peak in the pre-monsoon to monsoon window, whereas the UV index peaks in the dry pre-monsoon (March–April) and falls during the cloudy monsoon a distinct, radiation-driven seasonality.

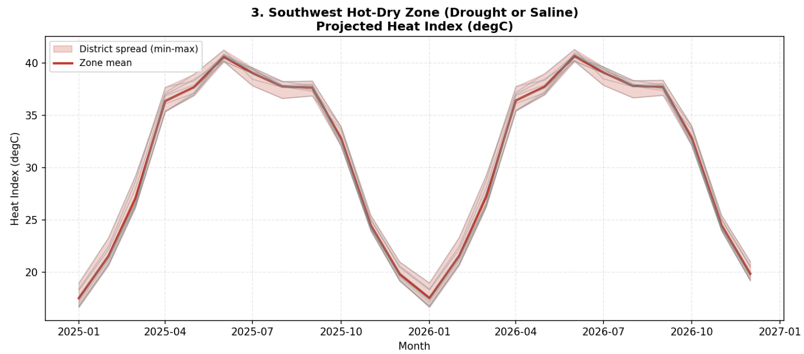

Figure 5: Projected heat index for the Southwest Hot–Dry Zone: the zone mean and the min–max envelope across member districts, illustrating within-zone spread.

Low class (79.8%). The UV signal was the most consistently hazardous condition: 75.5% of projected months fall in the WHO High class and a further 20.4% in Very High. Dew-point moisture stress was also severe, with 45.5% of months in the Extremely Oppressive class. Across zones, the Southwest Hot–Dry Zone carried the largest Extreme-Caution share for heat index (57.5%) and the only appearance of the Danger class (0.8%), whereas the Haor/Wetland Zone was the mildest (35.4% Extreme Caution); Figure 7 shows the full zonal composition and Figure 8 maps, for each district, the number of months per year spent at or above each condition's hazard threshold. The months-per-year view resolves the strong spatial and seasonal gradients that a dominant-class map obscures: the coastal south endures dew-point levels in the Extremely Oppressive class for roughly seven to eight months per year against three to four in the northwest, and very-high UV persists longest in the south.

Critically, the categorical layer validated well against observed 2025 (Table 6): exact-class agreement ranged from 81.5% (UV) to 89.8% (heat-index comfort), and agreement to within one adjacent class reached 100% for all

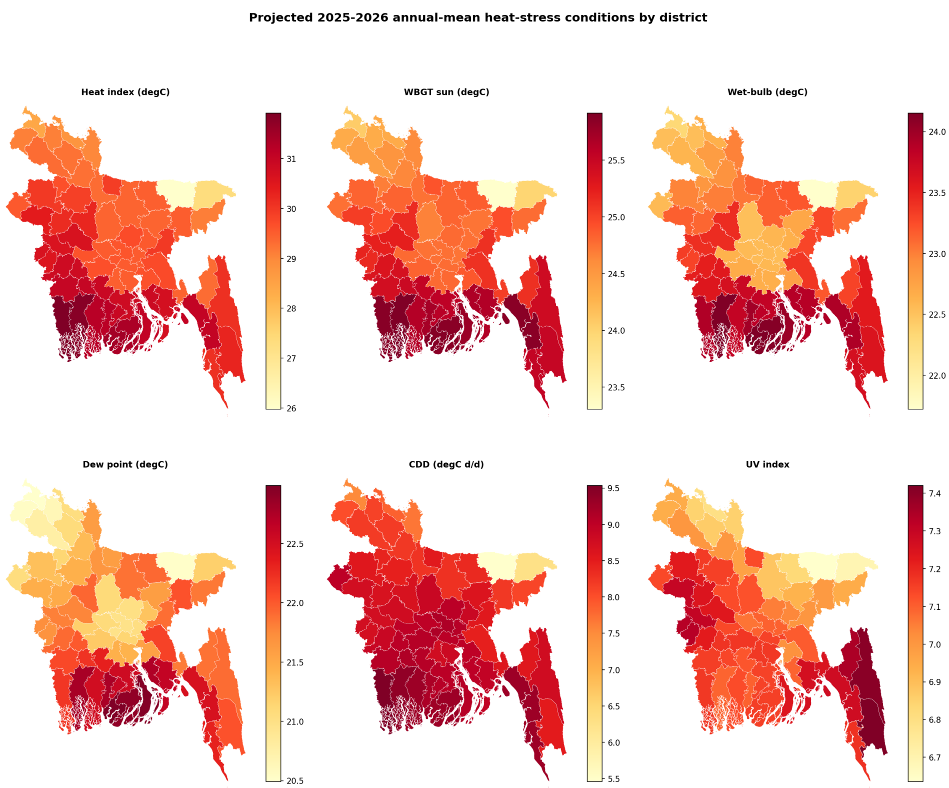

Figure 6: Projected 2025–2026 annual-mean heat-stress conditions by district: heat index, sun WBGT, wet-bulb temperature, dew point, cooling degree-days, and UV index. Every condition is shown at full district resolution rather than aggregated to zones.

five categorised conditions. The projection therefore reproduces not only the continuous fields but also their operational risk classification.

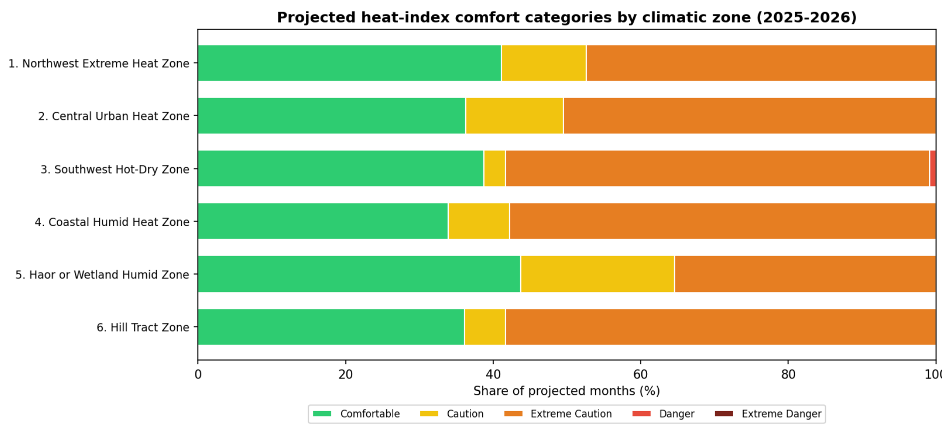

Figure 7: Projected heat-index comfort categories by climatic zone (share of projected months, 2025–2026). The Southwest Hot–Dry Zone carries the largest Extreme-Caution share and the only Danger occurrences; the Haor/Wetland Zone is mildest.

4.7. Projection uncertainty

The first-order error propagation yielded confidence bands that are approximately constant across the 24-month horizon, reflecting the dominance of the seasonal model's error over the small trend extrapolation. For the Dhaka exemplar, the 95% heat-index band was $\pm 1.17\,^{\circ}$C, the wet-bulb band $\pm 0.99\,^{\circ}$C, and the sun-WBGT band $\pm 0.76\,^{\circ}$C. These bands exceed the projected inter-annual trend signal by more than an order of magnitude, and this relationship is stated explicitly to caution against over-interpreting small year-to-year differences in the projection.

Table 5: Projected risk-category distribution (share of projected district-months, 2025–2026, 64 districts). Only populated categories are shown.

Condition	Category	Share (%)
Heat-index comfort	Comfortable	37.8
	Caution	10.2
	Extreme Caution	51.8
	Danger	0.1
WBGT (sun, outdoor)	Low	59.0
	Moderate	37.0
	High	3.9
WBGT (shade, indoor)	Low	79.8
	Moderate	20.2
UV (WHO)	Moderate	4.2
	High	75.5
	Very High	20.4
Dew-point moisture stress	Comfortable	19.7
	Slightly Humid	6.6
	Humid	16.3
	Very Humid / Oppressive	11.9
	Extremely Oppressive	45.5

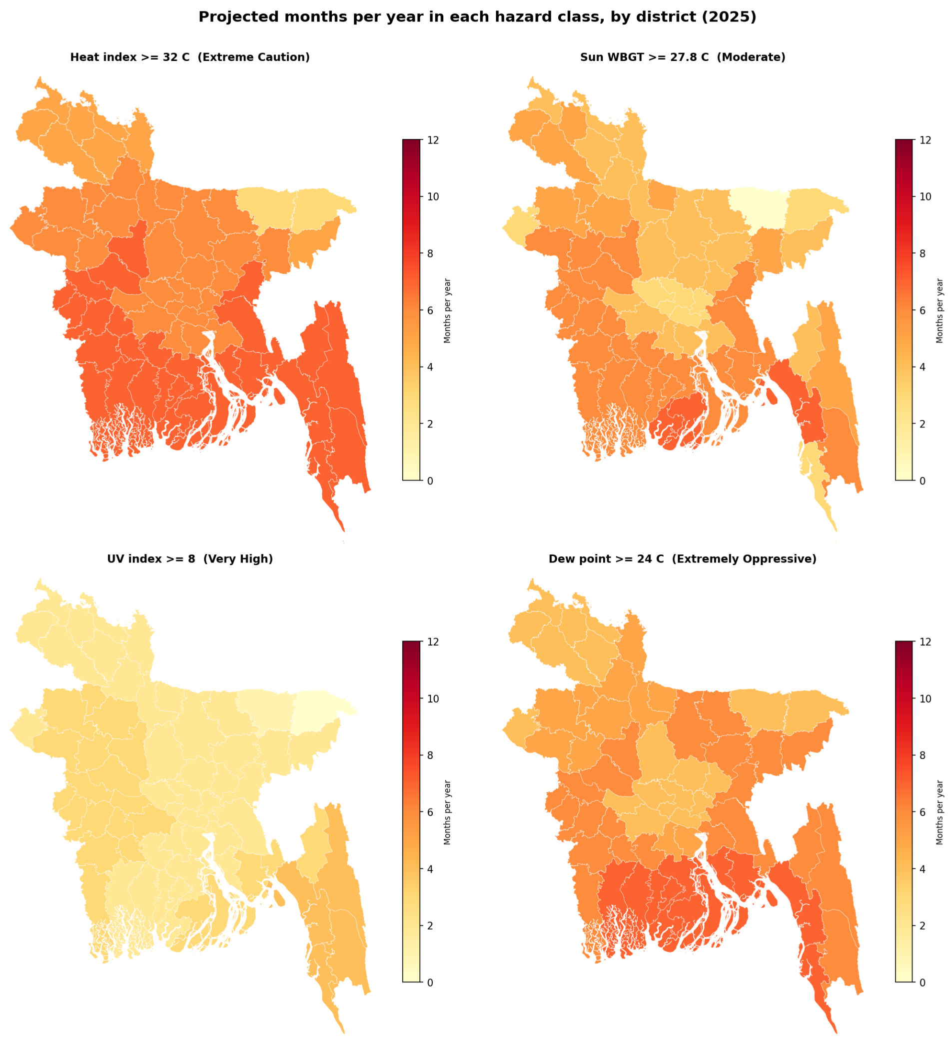

Figure 8: Projected number of months per year (of 12) that each district spends at or above a hazard threshold, for heat index ($\geq 32\,^{\circ}$C, Extreme Caution), sun WBGT ($\geq 27.8\,^{\circ}$C, Moderate), UV (≥ 8, Very High), and dew point ($\geq 24\,^{\circ}$C, Extremely Oppressive). This months-per-year measure preserves the seasonal and spatial structure that a single dominant-class map collapses.

Table 6: Categorical validation against observed 2025: exact-class agreement and agreement to within one adjacent class, pooled over district-months.

Condition	n	Exact-class (%)	Within one class (%)
Heat-index comfort	704	89.8	100.0
WBGT (sun, outdoor)	704	84.1	100.0
WBGT (shade, indoor)	704	88.8	100.0
UV (WHO)	704	81.5	100.0
Dew-point moisture stress	704	87.1	100.0

5. Discussion

5.1. Principal findings

This study delivers district-resolved, uncertainty-aware near-term projections of seven temperature-correlated heat-stress conditions for all 64 districts of Bangladesh, together with a leakage-free external validation against independent 2025 observations. The recomputed thermal indices validated strongly heat index, wet-bulb temperature, and WBGT all at median $R^2 \geq 0.95$ and, when mapped onto their operational risk classes, reproduced the observed 2025 class in 81–90% of district-months exactly and in 100% to within one adjacent class. Because the indices are computed from the projected primary variables rather than predicted directly, this accuracy is achieved without target leakage and while preserving the physical coupling among temperature, humidity, and radiation. This validated, multi-index projection extends prior Bangladeshi heat-projection work that has largely treated single indices in isolation most notably the heat-index appraisal of Rahman et al. [20] and the WBGT trend analysis of Kamal et al. [9] and complements single-algorithm forecasting precedents such as Binte Ahmed et al. [36] and Mohsin et al. [33] by covering all seven conditions jointly and validating them out-of-sample.

5.2. Skill must be interpreted correctly

A central and deliberately transparent result is that, although the ensembles comfortably outperform a seasonal-naive baseline, they are essentially at parity with monthly climatology (positive skill in only $\sim$32% of district–variable cases). This is expected rather than disappointing: at monthly resolution the tropical seasonal cycle dominates the variance, and any competent learner reconstructs the climatological annual march that climatology also encodes. The contribution of the framework is therefore not a reduction in monthly error but the provision of a spatially complete, uncertainty-quantified, and trend-consistent projection including physiologically relevant indices and their operational risk classes that a bare climatology cannot supply. We state this explicitly to discourage over-claiming predictive skill that the data do not support. This transparency deliberately contrasts with the common practice of reporting high absolute accuracy for instance the R^2 values near 0.99 and 0.88–0.97 reported for Bangladeshi heat-index and temperature prediction by Binte Ahmed et al. [36] and Mohsin et al. [33] without benchmarking against a climatological reference, which can make routine seasonal reconstruction appear as genuine forecasting skill.

5.3. Physical and spatial interpretation

The observational analysis reveals a coherent but spatially structured warming signal. Maximum temperature increased in every district (median $+0.56\,^{\circ}$C per decade) while minimum temperature remained essentially flat (median $+0.02\,^{\circ}$C per decade), implying a widening diurnal temperature range across much of the country a pattern with distinct implications for daytime heat exposure and nighttime physiological recovery. Humid-heat indices (heat index, wet-bulb, WBGT) rose in 57–58 of 64 districts, confirming that the warming is not purely a dry-bulb phenomenon. Spatially, the Southwest Hot–Dry and Coastal Humid Heat zones carry the greatest projected burden across perceived heat, humid heat, and cooling-energy demand, whereas the elevated Haor/Wetland basin is consistently mildest; the coastal belt in particular combines the highest wet-bulb temperatures and cooling degree-days, marking it as a priority region. These patterns matter physiologically: the elevated coastal wet-bulb burden erodes the evaporative margin that Raymond et al. [6] identified as the hard limit of human heat tolerance, in the same coastal belt where Ekra et al. [21] found the steepest historical rise in heat discomfort. The widening diurnal range is also epidemiologically salient, since reduced nighttime cooling impairs physiological recovery and compounds the heatstroke-mortality and labour-capacity risks documented for Bangladesh and for outdoor workers more broadly [26, 57].

5.4. Operational risk categories and their message

Translating the projections into established risk classes sharpens the public-health narrative. The UV signal is the most uniformly hazardous condition: essentially the entire country falls in the WHO High class or above throughout the projection, and dew-point moisture stress reaches the Extremely Oppressive class in almost half of all projected months. Heat-index comfort is dominated by the Extreme Caution class in the warm season, while the Danger class is essentially absent a direct and important consequence of categorising monthly means rather than daily maxima (Section 3.8), which the categories must be read against. The nationwide reach of the UV and humidity hazards, contrasted with the more spatially variable heat-index and WBGT burden, argues for differentiated adaptation responses: the near-universal High-to-Very-High UV exposure calls for the sun-protection measures codified in the WHO UV-index guidance [17], whereas the humid-heat burden is better addressed through the WBGT-based occupational-safety framework of ISO 7243 [4, 14], whose relevance is amplified by the large humid-heat labour losses estimated for South Asian outdoor workers [56].

5.5. Methodological reflections

Three design choices proved consequential. First, recomputing indices from projected primary variables (rather than modelling them directly) both prevents target leakage and guarantees internal physical consistency of the projected fields. Second, the detrend–residual–retrend construction resolves a failure mode of tree ensembles their inability to extrapolate a calendar-year feature so that projected years evolve with the observed climate trend instead of collapsing onto the last training year; imposing the robust 1980–2024 Sen slope additionally keeps the projection consistent with the manuscript's own trend analysis by construction. Third, the district choropleths exposed a data-quality problem (early-record missing values encoded as zero) that zonal-median summaries had masked, underscoring the value of full-resolution spatial diagnostics and prompting the physical-floor correction described in Section 3.12.

5.6. Limitations

Several limitations bound the interpretation. (i) The projection horizon is deliberately restricted to 24 months; because the re-added trend is small relative to the projection uncertainty (the 95% heat-index band exceeds the annual trend signal by more than an order of magnitude), longer horizons would add deterministic drift without validatable information. (ii) Working at monthly-mean resolution necessarily compresses daily extremes, so the risk categories describe typical monthly conditions and systematically under-represent short-duration dangerous events; a daily-resolution extension is the natural next step. (iii) A composite heatstroke-hazard index was implemented but excluded from predictive results for negative out-of-sample skill, and is reported only descriptively. (iv) All meteorological inputs derive from a single commercial reanalysis-blended source; despite strong 2025 validation, independent station-based corroboration would strengthen confidence, and the early-record inhomogeneity in a subset of districts (mitigated here by masking) illustrates the sensitivity of long-run trends to input quality. (v) The models add little skill over climatology at monthly scale, as discussed above. The 24-month horizon is complementary to, rather than competitive with, the multi-decadal CMIP6 projections available for Bangladesh and the wider region [58, 44]: those studies bound the long-term trajectory under emission scenarios, whereas the present framework supplies the near-term, district-resolved, externally validated detail that dynamical downscaling cannot yet deliver at this resolution.

5.7. Outlook

Several extensions follow naturally. A daily-resolution implementation would recover the short-duration extremes that monthly means smooth away and let the risk categories reflect daily peaks the regime in which the heatstroke indicator excluded here may regain skill [49]. Coupling the projected fields with health-surveillance and cooling-energy records would convert the physical outlook into quantified public-health and infrastructure burdens [26]. Methodologically, the framework can absorb recent advances in the individual conditions reanalysis-based UV retrieval [51] and attention-based dew-point models with explainability [55] and its near-term projections can be systematically benchmarked against CMIP6 dynamical downscaling [58] to bridge near-term operational forecasting and long-term climate projection.

6. Conclusion

This study introduced the Temperature-Correlated Conditions (TCC) framework and applied it to project seven physiologically and operationally relevant heat-stress conditions across all 64 districts of Bangladesh. By training gradient-boosting ensembles on the 2014–2024 record with future-known seasonal features, re-imposing the observed 1980–2024 trend, and recomputing every thermal index from the projected primary variables, the framework produced district-level 24-month projections that validated strongly against independent 2025 observations median R^2 of 0.95–0.98 for the key indices, with operational risk classes reproduced to within one category in 100% of district-months.

The observational analysis revealed a coherent but spatially structured warming signal: maximum temperature rose in every district (median $+0.56\,^{\circ}\mathrm{C}$ per decade) while minimum temperature remained essentially flat, implying a widening diurnal temperature range, and the humid-heat indices increased across most of the country. Spatially, the coastal and southwestern zones emerged as the areas of greatest combined perceived-heat, humid-heat, and cooling-energy burden, whereas ultraviolet and oppressive-humidity hazards were effectively nationwide. Crucially, the ensembles were found to be at parity with monthly climatology; the framework's contribution is therefore not improved monthly accuracy but a spatially complete, uncertainty-aware, and trend-consistent outlook delivered at full district resolution that supports targeted heat-adaptation planning. One indicator, heatstroke risk, was excluded from the predictive results for a lack of out-of-sample skill and reported only descriptively.

Several directions follow naturally. Extending the analysis to daily resolution would capture the short-duration extremes that monthly means smooth away and would allow the risk categories to reflect daily peaks rather than typical monthly conditions. Coupling the projected indices with health-surveillance and energy-demand records would translate this physical outlook into quantified public-health and infrastructure burdens. More broadly, the leakage-free, trend-consistent, and externally validated design demonstrated here offers a transferable template for district-level heat-stress projection in other data-constrained, climatically heterogeneous regions.

CRediT authorship contribution statement

Mohammad Khalid Hasan Nabil: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Project administration.

Tahia Parsha: Investigation, Software, Writing – original draft, Writing – review & editing.

Musayeb Hossain Usama: Investigation, Software, Writing – original draft, Writing – review & editing.

.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The raw daily meteorological data are available from the Visual Crossing Weather API (`https://www.visualcrossing.com`). The complete deterministic machine-learning pipeline source code, the interactive Streamlit web-application dashboard, and the comprehensive 64-district projected dataset are permanently archived on Zenodo (`https://doi.org/10.5281/zenodo.21418291`). The interactive dashboard—which visualises the projections, validation, and risk categories for all 64 districts and seven conditions—is served live at `https://tcc-64-districts-mohammad-khalid.streamlit.app/`.

References

[1] R. Shaw, Y. Luo, T. S. Cheong, S. Abdul Halim, S. Chaturvedi, M. Hashizume, G. E. Insarov, Y. Ishikawa, M. Jafari, A. Kitoh, J. Pulhin, C. Singh, K. Vasant, Z. Zhang, Asia, in: H.-O. Pörtner, D. C. Roberts, M. Tignor, E. S. Poloczanska, K. Mintenbeck, A. Alegría, M. Craig, S. Langsdorf, S. Löschke, V. Möller, A. Okem, B. Rama (Eds.), Climate Change 2022: Impacts, Adaptation and Vulnerability, Cambridge University Press, Cambridge, UK and New York, NY, USA, 2022, pp. 1457–1579. doi:10.1017/9781009325844.012.

[2] T. Kjellstrom, N. Maître, C. Saget, M. Otto, T. Karimova, Working on a Warmer Planet: The Impact of Heat Stress on Labour Productivity and Decent Work, Technical Report, International Labour Organization, Geneva, Switzerland, 2019. URL: https://www.ilo.org/publications/working-warmer-planet-impact-heat-stress-labour-productivity-and-decent.

[3] R. G. Steadman, The assessment of sultriness. Part I: A temperature-humidity index based on human physiology and clothing science, Journal of Applied Meteorology 18 (1979) 861–873. doi:10.1175/1520-0450(1979)018<0861:TAOSPI>2.0.CO;2.

[4] International Organization for Standardization, Ergonomics of the thermal environment Assessment of heat stress using the WBGT (wet bulb globe temperature) index, 2017.

[5] J. Spinoni, J. V. Vogt, P. Barbosa, A. Dosio, N. McCormick, A. Bigano, H.-M. Füssel, Changes of heating and cooling degree-days in Europe from 1981 to 2100, International Journal of Climatology 38 (2018) e191–e208. doi:10.1002/joc.5362.

[6] C. Raymond, T. Matthews, R. M. Horton, The emergence of heat and humidity too severe for human tolerance, Science Advances 6 (2020) eaaw1838. doi:10.1126/sciadv.aaw1838.

[7] A. Dewan, G. Kiselev, D. Botje, Diurnal and seasonal trends and associated determinants of surface urban heat islands in large Bangladesh cities, Applied Geography 135 (2021) 102533. doi:10.1016/j.apgeog.2021.102533.

[8] R. Abrar, S. K. Sarkar, K. T. Nishtha, S. Talukdar, Shahfahad, A. Rahman, A. R. M. T. Islam, A. Mosavi, Assessing the spatial mapping of heat vulnerability under urban heat island (uhi) effect in the dhaka metropolitan area, Sustainability 14 (2022) 4945. doi:10.3390/su14094945.

[9] A. S. M. M. Kamal, A. K. F. Fahim, S. Shahid, Changes in wet bulb globe temperature and risk to heat-related hazards in Bangladesh, Scientific Reports 14 (2024) 10417. doi:10.1038/s41598-024-61138-8.

[10] L. Breiman, Random forests, Machine Learning 45 (2001) 5–32.

[11] C.-L. Hwang, K. Yoon, Multiple Attribute Decision Making: Methods and Applications, Springer, 1981.

[12] L. P. Rothfusz, The Heat Index Equation (or, More Than You Ever Wanted to Know About Heat Index), Technical Report SR/SSD 90-23, NWS Southern Region Headquarters, Fort Worth, TX, 1990.

[13] C. P. Yaglou, D. Minard, Control of heat casualties at military training centers, AMA Archives of Industrial Health 16 (1957) 302–316.

[14] K. C. Parsons, Heat stress standard ISO 7243 and its global application, Industrial Health 44 (2006) 368–379. doi:10.2486/indhealth.44.368.

[15] R. Stull, Wet-bulb temperature from relative humidity and air temperature, Journal of Applied Meteorology and Climatology 50 (2011) 2267–2269. doi:10.1175/JAMC-D-11-0143.1.

[16] M. Bhatnagar, J. Mathur, V. Garg, Determining base temperature for heating and cooling degree-days for India, Journal of Building Engineering 18 (2018) 270–280. doi:10.1016/j.jobe.2018.03.020.

[17] World Health Organization, World Meteorological Organization, United Nations Environment Programme, International Commission on Non-Ionizing Radiation Protection, Global Solar UV Index: A Practical Guide, World Health Organization, Geneva, 2002.

[18] M. G. Lawrence, The relationship between relative humidity and the dewpoint temperature in moist air: A simple conversion and applications, Bulletin of the American Meteorological Society 86 (2005) 225–233. doi:10.1175/BAMS-86-2-225.

[19] H.-O. Pörtner, D. C. Roberts, H. Adams, C. Adler, P. Aldunce, E. Ali, R. A. Begum, R. B. Betts, R. Bezner Kerr, R. Biesbroek, J. Birkmann, K. Bowen, E. Castellanos, G. Cissé, A. Constable, W. Cramer, D. Dodman, S. H. Eriksen, A. Fischlin, M. Garschagen, B. Glavovic, E. Gilmore, M. Haasnoot, S. Harper, T. Hasegawa, B. Hayward, Y. Hirabayashi, M. Howden, K. Kalaba, W. Kiessling, R. Lasco, J. Lawrence, M. F. Lemos, R. Lempert, D. L. Ley, T. Lissner, S. Lluch-Cota, S. Loeschke, S. Lucatello, Y. Luo, B. Mackey, S. Maharaj, C. Mendez, K. Mintenbeck, V. Möller, M. M. Vale, M. D. Morecroft, A. Mukherji, M. Mycoo, T. Mustonen, J. Nalau, A. Okem, J. P. Ometto, C. Parmesan, M. Pelling, P. Pinho, E. S. Poloczanska, M.-F. Racault, D. Reckien, J. Pereira, A. Revi, S. Rose, R. Sanchez-Rodriguez, E. L. F. Schipper, D. Schmidt, D. Schoeman, R. Shaw, C. Singh, W. Solecki, L. Stringer, A. Thomas, E. Totin, C. Trisos, M. van Aalst, D. Viner, M. Wairiu, R. Warren, P. Yanda, Z. Z. Ibrahim, Summary for policymakers, in: H.-O. Pörtner, D. C. Roberts, M. Tignor, E. S. Poloczanska, K. Mintenbeck, A. Alegría, M. Craig, S. Langsdorf, S. Löschke, V. Möller, A. Okem, B. Rama (Eds.), Cli-

mate Change 2022: Impacts, Adaptation and Vulnerability. Contribution of Working Group II to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change, Cambridge University Press, Cambridge, UK and New York, NY, USA, 2022, pp. 3–33. doi:https://doi.org/10.1017/9781009325844.001.

[20] M. B. Rahman, R. Salam, A. R. M. T. Islam, A. Tasnuva, U. Haque, S. Shahid, Z. Hu, J. Mallick, Appraising the historical and projected spatiotemporal changes in the heat index in Bangladesh, Theoretical and Applied Climatology 146 (2021) 125–138. doi:10.1007/s00704-021-03705-x.

[21] A. T. Ekra, M. M. Hamed, Z. Ali, M. K. I. B. Muhammad, M. M. H. Khan, M. Kamruzzaman, S. Shahid, Changes in human heat discomfort and its drivers in bangladesh, Urban Climate 55 (2024) 101884. URL: https://www.sciencedirect.com/science/article/pii/S2212095524000804. doi:10.1016/j.uclim.2024.101884.

[22] A. S. M. M. Kamal, A. K. F. Fahim, S. Shahid, Simplified equations for wet bulb globe temperature estimation in Bangladesh, International Journal of Climatology 44 (2024) 1636–1653. doi:10.1002/joc.8402.

[23] I. Ahmed, S. Ishtiaque, T. Zahan, et al., Climate change vulnerability in Bangladesh based on trend analysis of some extreme temperature indices, Theoretical and Applied Climatology 149 (2022) 831–842. doi:10.1007/s00704-022-04079-4.

[24] A. R. M. T. Islam, I. Ahmed, M. S. Rahman, et al., Spatiotemporal trends of temperature extremes in Bangladesh under changing climate

using multi-statistical techniques, Theoretical and Applied Climatology 147 (2021) 137–152. doi:10.1007/s00704-021-03828-1.

[25] A. R. M. T. Islam, I. Ahmed, M. S. Rahman, Trends in cooling and heating degree-days overtimes in Bangladesh? An investigation of the possible causes of changes, Natural Hazards 101 (2020) 879–909. doi:10.1007/s11069-020-03900-5.

[26] M. A. Miah, et al., Investigating the association between heatstroke mortality and climatic factors in Bangladesh: An ecological time series study, Health Science Reports (2026). doi:10.1002/hsr2.72510.

[27] A. Dewan, G. Kiselev, D. Botje, G. I. Mahmud, M. H. Bhuian, Q. K. Hassan, Diurnal and seasonal trends and associated determinants of surface urban heat islands in large Bangladesh cities, Urban Climate 38 (2021) 100896. doi:10.1016/j.uclim.2021.100896.

[28] A.-A. Faisal, A.-A. Kafy, A. Al Rakib, et al., Impact of urbanization on urban heat island intensity in major districts of Bangladesh using remote sensing and geo-spatial tools, Climate 10 (2022) 3. doi:10.3390/cli10010003.

[29] S. Park, et al., Characteristics of the urban heat island in Dhaka, Bangladesh, and its interaction with heat waves, Asia-Pacific Journal of Atmospheric Sciences (2024). doi:10.1007/s13143-024-00362-8.

[30] T. Chen, C. Guestrin, Xgboost: A scalable tree boosting system, in: Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, Association for Comput-

ing Machinery, New York, NY, United States, 2016, pp. 785–794. doi:https://doi.org/10.1145/2939672.2939785.

[31] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, T.-Y. Liu, Lightgbm: A highly efficient gradient boosting decision tree, in: I. Guyon, U. V. Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, R. Garnett (Eds.), Advances in Neural Information Processing Systems 30, Curran Associates, Inc., 2017, pp. 3146–3154. URL: https://proceedings.neurips.cc/paper/2017/hash/6449f44a102fde848669bdd9eb6b76fa-Abstract.html.

[32] L. Prokhorenkova, G. Gusev, A. Vorobev, A. V. Dorogush, A. Gulin, Catboost: unbiased boosting with categorical features, in: S. Bengio, H. Wallach, H. Larochelle, K. Grauman, N. Cesa-Bianchi, R. Garnett (Eds.), Advances in Neural Information Processing Systems 31, Curran Associates, Inc., 2018, pp. 6639–6649. URL: https://proceedings.neurips.cc/paper/2018/hash/14491b756b3a51daac41c24863285549-Abstract.html.

[33] M. Mohsin, T. Ghosh, F. Akter, S. Sarkar, M. R. A. Mullick, Seasonal weather pattern prediction from ENSO indices using machine learning, Discover Environment 4 (2026) 29. doi:10.1007/s44274-026-00533-6.

[34] M. R. Akter, M. H. Rahman, Analysis of climatic factors and utilization of machine learning techniques to anticipate humidity levels in northern Bangladesh, American Journal of Data Mining and Knowledge Discovery 10 (2025) 1–20. doi:10.11648/j.ajdmkd.20251001.11.

[35] M. H. Rahman, et al., Comparison of RNN-LSTM, TFDF and stacking model approach for weather forecasting in Bangladesh using historical

data from 1963 to 2022, PLOS ONE (2024). doi:10.1371/journal.pone.0305406.

[36] A. Binte Ahmed, Z. Zijan, A. Joy, R. Rois, Machine-learning framework for conditional estimation and scenario-based projection of the heat index for public health interventions, PLOS One 21 (2026). doi:10.1371/journal.pone.0348202.

[37] H.-j. Kim, et al., Ensemble machine learning of gradient boosting (XGBoost, LightGBM, CatBoost) and attention-based CNN-LSTM for harmful algal blooms forecasting, Toxins 15 (2023) 608. doi:10.3390/toxins15100608.

[38] F. J. Galán-Sales, et al., An approach to enhance time series forecasting by fast Fourier transform, in: Proceedings of the 18th International Conference on Soft Computing Models in Industrial and Environmental Applications (SOCO 2023), Springer, 2024. doi:10.1007/978-3-031-42529-5_25.

[39] M. S. Hanoon, A. N. Ahmed, N. Zaini, A. Razzaq, P. Kumar, M. Sherif, A. Sefelnasr, A. El-Shafie, Developing machine learning algorithms for meteorological temperature and humidity forecasting at terengganu state in malaysia, Scientific Reports 11 (2021) 18935. doi:https://doi.org/10.1038/s41598-021-96872-w.

[40] M. Abdelsattar, A. AbdelMoety, A. Emad-Eldeen, Comparative analysis of machine learning techniques for temperature and humidity prediction in photovoltaic environments, Scientific Reports 15 (2025) 15650. doi:https://doi.org/10.1038/s41598-025-98607-7.

[41] O. C. Pasche, J. Wider, Z. Zhang, J. Zscheischler, S. Engelke, Validating deep learning weather forecast models on recent high-impact extreme events, Artificial Intelligence for the Earth Systems 4 (2025). doi:https://doi.org/10.1175/AIES-D-24-0033.1.

[42] D. I. Jeong, B. Yu, A. J. Cannon, Comparison of two distinct leading modes in the variability of summer humidex and temperature heatwaves over north america, Climate Dynamics 63 (2025) 1457–1579. URL: https://link.springer.com/article/10.1007/s00382-025-07669-w. doi:https://doi.org/10.1007/s00382-025-07669-w.

[43] J. Sojan, J. Srinivasan, A comparative analysis of accelerating humid and dry heat stress in india, Environmental Research Communications 6 (2024). doi:10.1088/2515-7620/ad2490.

[44] R. Kushwaha, P. Kumar, Y. Hisaki, Global future heat stress projections: Regional variations of humidex changes from high-resolution cmip6 models, Atmospheric Research 327 (2026) 108367. URL: https://www.sciencedirect.com/science/article/pii/S0169809525004594. doi:https://doi.org/10.1016/j.atmosres.2025.108367.

[45] E. Scoccimarro, P. G. Fogli, S. Gualdi, The role of humidity in determining scenarios of perceived temperature extremes in europe, Environmental Research Letters 12 (2017) 114029. doi:10.1088/1748-9326/aa8cdd.

[46] P. Saha, R. Mahanta, Y. Yamane, F. Murata, H. Kamimera, T. Terao, Post-2000 Escalation of Humid Heatwaves and Emerging Cloudburst

Risks across the South Asia, Asian Journal of Engineering Geology 2 (2025) 157–158. doi:10.64862/ajeg.2025.2sp.73.197.

[47] A. Tabassum, K. Park, S.-H. Hong, J.-J. Baik, H. Beom-Soon, Impacts of cool roofs on urban heat island and air quality in dhaka, bangladesh: A case modeling study during a heat wave, Atmospheric Pollution Research 16 (2025) 102549. doi:10.1016/j.apr.2025.102549.

[48] H. Xu, S. Guo, X. Shi, Y. Wu, J. Pan, H. Gao, Y. Tang, A. Han, Machine learning-based analysis and prediction of meteorological factors and urban heatstroke diseases, Frontiers in Public Health 12 (2024) 1420608. doi:https://doi.org/10.3389/fpubh.2024.1420608.

[49] T. Ehsan, F. A. Bina, G. M. H. Bashar, L. C. Paul, A machine learning-based forecasting approaches and correlation analysis to assess future extreme heat scenarios in bangladesh and its impact on public health, Engineering Reports 8 (2026) e70588. doi:https://doi.org/10.1002/eng2.70588.

[50] M. Bilgili, Time series forecasting on cooling degree-days (cdd) using sarima model, Natural Hazards 118 (2023) 2569–2592. doi:https://doi.org/10.1007/s11069-023-06109-4.

[51] S. Teggi, S. Costanzini, F. Despini, et al., Uv index from era5 reanalysis, Scientific Reports 16 (2026) 12950. doi:https://doi.org/10.1038/s41598-026-40878-9.

[52] K. Mohammadi, S. Shamshirband, S. Motamedi, D. Petković, R. Hashim, M. Gocic, Extreme learning machine based prediction of daily dew point temperature, Computers and Elec-

tronics in Agriculture 117 (2015) 214–225. URL: https://www.sciencedirect.com/science/article/pii/S0168169915002343. doi:https://doi.org/10.1016/j.compag.2015.08.008.

[53] S. N. Qasem, S. Samadianfard, H. Sadri Nahand, A. Mosavi, S. Shamshirband, K.-w. Chau, Estimating daily dew point temperature using machine learning algorithms, Water 11 (2019) 582. doi:https://doi.org/10.3390/w11030582.

[54] F. Yao, J. Sun, J. Dong, Estimating daily dew point temperature based on local and cross-station meteorological data using catboost algorithm, CMES - Computer Modeling in Engineering and Sciences 130 (2021) 671–700. URL: https://www.sciencedirect.com/science/article/pii/S1526149221000588. doi:https://doi.org/10.32604/cmes.2022.018450.

[55] M. Abdullah, S. Waheed, M. Mahmodul Hasan, M. Morshed, Explainable ai-driven dew point forecasting with attention-based temporal convolutional networks, IEEE Access 13 (2025) 59923–59948. doi:https://doi.org/10.1109/ACCESS.2025.3557263.

[56] L. A. Parsons, Y. J. Masuda, T. Kroeger, D. Shindell, N. H. Wolff, J. T. Spector, Global labor loss due to humid heat exposure underestimated for outdoor workers, Environmental Research Letters 17 (2021) 014050. doi:10.1088/1748-9326/ac3dae.

[57] L. G. Ioannou, J. Foster, N. B. Morris, J. F. Piil, G. Havenith, I. B. Mekjavic, G. P. Kenny, L. Nybo, A. D. Flouris, Occupational heat strain in outdoor workers: A comprehensive review and meta-analysis, Tem-

perature 9 (2022) 67–102. doi:https://doi.org/10.1080/23328940.2022.2030634.

[58] M. K. Khatun, et al., Temperature extremes projections over Bangladesh from CMIP6 multi-model ensemble, Theoretical and Applied Climatology (2024). doi:10.1007/s00704-024-05173-5.

[59] FAO/UNDP, Land Resources Appraisal of Bangladesh for Agricultural Development, Technical Report, Food and Agriculture Organization of the United Nations, Rome, Italy, 1988.

[60] H. E. Rashid, Geography of Bangladesh, University Press Limited, Dhaka, Bangladesh, 1991.

[61] Visual Crossing Corporation, Visual crossing weather data, https://www.visualcrossing.com/, 2025. Accessed: 2026-07-16.

[62] D. Runfola, A. Anderson, H. Baier, et al., geoboundaries: A global database of political administrative boundaries, PLOS ONE 15 (2020) e0231866.

[63] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, et al., Scikit-learn: Machine learning in Python, Journal of Machine Learning Research 12 (2011) 2825–2830.